The Celestial Sphere, Coordinates and Constellation History

Covers Lectures 2.1–2.3 about 75 minutes

Name: _____ Date: _____ Marks: _____ / 75

You may use: a calculator. Do Sections A–D from memory first. **Difficulty:** ★ remember ★★ apply ★★★ challenge

Section A — Fill in the blanks ★ 10 marks

1. The point directly overhead is the _____; the point directly below is the _____.
2. The _____ is the great circle passing through the North point, the zenith and the South point.
3. The altitude of the north celestial pole equals the observer's _____.
4. The celestial equator always meets the horizon at the _____ and _____ points.
5. In right ascension, 1 hour = _____ degrees.
6. Azimuth is measured along the horizon starting from _____, going towards _____.
7. The IAU recognises _____ constellations, adopted in the year _____.
8. Ptolemy's book, the _____, listed _____ constellations.
9. Johann Bayer labelled stars with _____ letters in his atlas *Uranometria* (1603).
10. The smallest constellation is _____ and the largest is _____.

Section B — Multiple choice ★ 10 marks

1. Which coordinates of a star change from minute to minute? (a) RA and Dec (b) altitude and azimuth (c) ecliptic latitude (d) none
2. A star with Dec = 0° rises: (a) due north (b) due east (c) north-east (d) south-east
3. A star at azimuth 225° is in the: (a) north-east (b) south-east (c) south-west (d) north-west
4. From Delhi (28.6° N) a star is circumpolar if its declination is greater than: (a) +28.6° (b) +61.4° (c) +90° (d) −61.4°
5. "α Centauri" means: (a) the (usually) brightest star of Centaurus (b) the first star discovered in Centaurus (c) the nearest star (d) the central star
6. The southern constellations named after instruments (Telescopium, Microscopium…) were introduced by: (a) Ptolemy (b) Hevelius (c) Lacaille (d) Bayer
7. The precise boundaries of the 88 constellations were drawn by: (a) Eugène Delporte (b) Galileo (c) Hipparchus (d) Charles Messier
8. A star is on your meridian when the local sidereal time equals its: (a) declination (b) right ascension (c) azimuth (d) altitude
9. Which of these is an ASTERISM, not a constellation? (a) Orion (b) Lyra (c) the Saptarishi (Big Dipper) (d) Crux
10. The constellation **Indus** ("the Indian") was introduced by: (a) Indian astronomers (b) Dutch navigators in the 1590s (c) the IAU in 1922 (d) Lacaille

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Section C — Match the astronomer ★ 8 marks

Person	Contribution
1. Ptolemy	A. Constellation boundaries (1930)
2. Al-Sufi	B. Numbered stars from west to east
3. Lacaille	C. *Almagest*: 48 constellations, 1,022 stars
4. Hevelius	D. *Book of Fixed Stars* (964 CE); first description of the Andromeda Galaxy
5. Bayer	E. Greek-letter star names (1603)
6. Delporte	F. Discovered precession; graded stars by brightness
7. Hipparchus	G. Lynx, Lacerta, Scutum, Sextans
8. Flamsteed	H. Telescopium, Antlia, Octans; split Argo Navis

Answers: 1–____ 2–____ 3–____ 4–____ 5–____ 6–____ 7–____ 8–____

Section D — Indian sky names ★ 8 marks

Write the modern name of each star or star group.

Indian name	Modern name	Indian name	Modern name
1. Dhruva		5. Rohini	
2. Lubdhaka		6. Krittika	
3. Agastya		7. Saptarishi	
4. Trishanku		8. Arundhati	

Section E — Calculations ★★ 29 marks

Formula box

$1^{h} = 15^{\circ}$, $1^{m} = 15'$, $1^{s} = 15''$ $h_{\text{transit}} = 90^{\circ} - \varphi + \delta$ (if $> 90^{\circ}$: north of the zenith at $180^{\circ} - h$)
Circumpolar: $\delta > 90^{\circ} - \varphi$; never rises: $\delta < -(90^{\circ} - \varphi)$ Small angle: $\theta(^{\circ}) = 57.3\,D/d$
LST $\approx \alpha_{\odot}$ + (local time $- 12^{h}$), $\alpha_{\odot} \approx 0^{h}$ on 21 March, $+2^{h}$ per month Hand: finger 1°, three fingers 5°, fist 10°

Worked example

Meridian altitude of Aldebaran ($\delta = +16.5^{\circ}$) from Kolkata ($\varphi = 22.6^{\circ}$): $90 - 22.6 + 16.5 = \mathbf{83.9}^{\circ}$, in the south — almost overhead.

E1. [4] Convert: (a) RA $5^{h}30^{m}$ to degrees (b) 270° to RA (c) RA $13^{h}25^{m}$ to degrees (d) $3^{\circ}15'$ to decimal degrees.

E2. [6] Find the transit altitude and direction (N or S of the zenith): (a) Antares ($\delta = -26.4^{\circ}$) from Delhi (28.6° N); (b) Vega ($\delta = +38.8^{\circ}$) from Mumbai (19.1° N); (c) Canopus ($\delta = -52.7^{\circ}$) from Chennai (13.1° N).

E3. [6] Fill in **C** (circumpolar), **R** (rises and sets) or **N** (never rises):

Star	Dec	Bengaluru (13° N)	Srinagar (34° N)
Polaris	+89.3°		
Kochab	+74.2°		
Dubhe	+61.8°		
Capella	+46.0°		
Canopus	−52.7°		
Acrux	−63.1°		

E4. [3] Dubhe and Merak are 5.4° apart and Polaris lies about 5 times this distance from Dubhe. (a) How many degrees? (b) About how many fists? (c) Two stars are "2 fists + 3 fingers" apart — what is their separation?

E5. [4] (a) The Sun is 13,92,000 km wide and 15 crore km away. Find its angular size. (b) The ISS is 109 m long and ∼400 km away when overhead. Find its angular size in arcseconds. Could you see its shape with the naked eye (resolution ≈ 60″)?

E6. [6] (a) Estimate the LST at 10 PM (local time) on 21 November. (b) The Pleiades have RA 3.8^{h}. Are they east or west of the meridian then, and how long until they cross it? (c) In which month is Arcturus (RA 14.3^{h}) on the meridian at 9 PM?

Section F — Short answers ★★ 10 marks

1. Why do astronomers measure right ascension in **hours** rather than degrees?
2. Why are the stars of a constellation usually **not** physically connected? Give one example from Orion.
3. Why are RA–Dec coordinates printed in star catalogues, while alt–az are not?
4. Why can people in Kerala see the Southern Cross, while people in Srinagar cannot?
5. A friend says "Polaris is the brightest star in the sky." Correct them and explain why Polaris is nevertheless important.

Bonus ★★★ up to 6 marks

1. (3) At which latitude would **Sirius** ($\delta = -16.7°$) pass exactly through the zenith? Which hemisphere? Name a country there.
2. (3) Show that, from the **equator**, every star rises and sets and is above the horizon for exactly 12 hours.